The effect of oral alpha-galactosidase on intestinal gas production and gas-related symptoms.

Bloating, abdominal distention, and flatulence represent very frequent complaints in functional disorders but their pathophysiology and treatment are largely unknown. Patients frequently associate these symptoms with excessive intestinal gas and the reduction of gas production may represent an effective strategy. The aim was to evaluate the effect of alpha-galactosidase administration, in a randomized double-blind placebo-controlled protocol, on intestinal gas production and gas-related symptoms after a challenge test meal in healthy volunteers. Eight healthy volunteers ingested 300 or 1200 GalU of alpha-galactosidase or placebo during a test meal containing 420 g of cooked beans. Breath hydrogen excretion and occurrence of bloating, abdominal pain, discomfort, flatulence, and diarrhea were measured for 8 hr. The administration of 1200 GalU of alpha-galactosidase induced a significant reduction of both breath hydrogen excretion and severity of flatulence. A reduction in severity was apparent for all considered symptoms, but both 300 and 1200 GalU induced a significant reduction in the total symptom score. Alpha-galactosidase reduced gas production following a meal rich in fermentable carbohydrates and may be helpful in patients with gas-related symptoms.